KEL225

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Teaching Note:
Ariba Implementation at MED-X

Synopsis

The Ariba Implementation at MED-X case is designed to teach students how to analyze a program that is experiencing problems and recommend solutions. Specifically, the case introduces students to earned value analysis and program oversight for an e-procurement technology program. The case centers on MED-X's need to quickly discover why the company's e-procurement implementation project was not going according to plan. Once a cause has been discovered students will need to make a recommendation to fix the problem. Data for the simplified program, consisting of two concurrent projects, is given to students in the case, who should in turn analyze the project using earned value analysis. The case is an easy introduction to program management and oversight for executives and MBA students, and teaches the essentials of earned value project management. This case and the associated teaching note are used as an introduction to project management in the Kellogg MBA course entitled *Managing Technology Portfolios and Projects* and in multiple Kellogg executive programs, including *Driving Strategic Results through IT Portfolio Management*.

Supplementary Materials

FOR INSTRUCTORS

The following materials have been provided with this teaching note to aid instructors in leading this case:

- *Ariba Implementation at MED-X Debrief Slides* (PowerPoint)
- *Ariba Implementation at MED-X Solution* (Excel)

FOR STUDENTS

The following may be used as supplemental material to accompany the case:

- *Ariba Implementation at MED-X Exhibits* (Excel)

Purpose

The case centers on MED-X's need to quickly discover why its e-procurement implementation project was not going according to plan. For the purpose of the case, the complex program was simplified to two parallel projects: a software project and a hardware infrastructure project. Earned value data for the two projects are given to students, who should in turn analyze the plan using Excel and make recommendations. The learning from this simplified case is applicable to very complex programs.

Case Questions

The case assignment is as follows:

Your team has been assigned to help Christopher Martin determine what went wrong with the project and develop a presentation and recommendation for Terry Baker. Time is short, but a copy of an earned value analysis template has been provided to help you. This template is included in Case Exhibit 8, and is also available in the Excel file accompanying the case.

Additional related questions are:

1. *Which of the two components are underperforming according to the plan? How do you know this?*
2. *Are the components of the project within budget? How do you know?*
3. *What can you conclude by looking at the combined earned value data for the project?*
4. *Why did Terry Baker think that the project was going according to plan the entire time?*
5. *How much longer will the project take?*
6. *What should Martin have done earlier in the project timeline to prevent delays?*
7. *What should Martin do when managing future projects to prevent similar problems from developing?*

An additional disclaimer is given to students, who may only focus on the answers to these questions: *These questions are intended to help your analysis. However, your final solution to the case and recommendations to management should not necessarily be limited to the answers to these questions or the assumptions in the case.*

Analysis

This case was designed to be a self-contained introduction to earned value analysis (EVA). The analysis involves looking at the project plan and determining the best course of action.

Students will also quantitatively determine the project completion date and costs. They should use the EVA spreadsheet templates so their basic analysis follows a simple format.

INTRODUCTION TO EARNED VALUE PROGRAM MANAGEMENT

A major challenge for managers is to track the progress of IT projects. This can be difficult when the number of projects for a large firm may exceed one hundred. Earned value program management is a systematic approach to integrating the measurement of costs, schedule, and scope. It also allows managers to quickly determine the status of projects under their supervision.

Earned value program management allows the manager to answer the following questions objectively: Where have we been? Where are we now? Where are we going? Traditional forms of program management track only two data sources: budgeted costs and actual costs. By comparing these two items, one can only determine what has been spent versus what was planned to be spent. The method does not take into account what has actually been produced. Thus, one cannot determine what has been produced for the money spent or if the project is on schedule. In essence, the true performance of the project cannot be measured using the traditional methods.

In earned value program management there are three data sources: the budgeted cost of work scheduled, the actual cost of work scheduled, and the budgeted cost of work performed or the earned value. This technique allows managers to determine the schedule and cost variance for the project. Both the traditional method and the earned value method show the manager how much money was spent at any given time during the project. However, earned value provides the manager with additional key pieces of information: first, how much work has been completed to date for the funds spend, and second, what the total job cost will be and how long it will take to complete the job. The key obstacle in implementing earned value program management is measuring what has been accomplished, or the budgeted cost of work performed. Many organizations are not set up to record this critical piece of information in a timely manner. The implementation of a reliable and accurate system of recording the budgeted cost of work performed is an absolute must for the use of earned value program management.

In order to understand EVA, it is important to understand its terminology. Below is a list of key terms used in EVA:

- *Budgeted Cost of Work Scheduled (BCWS)*: the planned work for the project
- *Actual Cost of Work Performed (ACWP)*: the actual cost of the work done on the project
- *Budgeted Cost of Work Performed (BCWP)*: the earned value for the project; the actual dollar value of the work done relative to the original plan
- *Cost Variance (CV)*: the actual cost variance relative to earned value

  CV = BCWP – ACWP

- *Cost Performance Index (CPI)*: the ratio of the actual cost of work performed relative to earned value

  CPI = BCWP/ACWP

- *Schedule Variance (SV)*: the earned value schedule variance relative to the original plan

  SV = BCWP – BCWS

- *Schedule Performance Index (SPI)*: the ratio of planned work to earned value

  SPI = BCWP/BCWS

- *Control Ratio (CR)*: the indicator of the combined impact of schedule and cost variances relative to earned value

  CR = SPI × CPI

Exhibit TN1 illustrates some key EVA terms in the form of a graph. The graph exhibits a scenario in which a project is over budget but on time and delivery schedule. The actual cost (ACWP) line, in red, is above the baseline curve. The budgeted cost (BCWP) or earned value line, in blue, is below the baseline curve. This indicates the project has a cost overrun and a labor shortfall.

Exhibit TN2 is an overview of a project that is useful for high-level managers. By using the critical ratio, high-level executives can quickly determine how projects are performing. The control limits help identify what actions need to be taken, if necessary. Any project will have normal fluctuations in its critical ratio over time, but keeping a vigilant eye on the critical ratio can rescue a project before it is too late.[1]

FRAMEWORK FOR SOLVING THE CASE

Students will work through the case in groups. They will likely approach the case by entering the information about the project into project-planning software, such as Microsoft Project. While it is possible to complete the case without the aid of project-planning software, the exercise becomes much more tedious. Thus, learning to use this software is really a part of the case. From the software, students can easily tabulate metrics and costs, make "what-if" scenario changes, and export data into a spreadsheet to calculate the earned value information.

DETAILED SOLUTION

Step 1: Updating the EVA Template

The first task for the students will be to take the data from Exhibits 5, 6, and 7 from the case and insert it into the EVA template (Exhibit 8 from the case). If the data are entered correctly, the students' resulting figures should be identical to those presented at the end of this teaching note (see **Exhibit TN3** through **Exhibit TN5**).

Step 2: Analyzing the EVA Template

To better interpret the results of the EVA, students may choose to graph the cumulative spending over time for the software project, technical infrastructure project, and/or the combined program. If the data are entered correctly, the students' resulting graphs should be identical to those presented at the end of this teaching note (see **Exhibit TN6** through **Exhibit TN8**). Exhibit TN1 provides a generic example interpreting a sample graph and may help the students better understand EVA.

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[1] Additional detailed information regarding EVA can be found in Chapter 8 of James P. Lewis, *Fundamentals of Project Management* (New York: AMACOM, 2001).

Referring back to the results of the EVA template (Exhibits TN3, TN4, and TN5) the control ratio, also called the critical ratio (CR), is helpful in determining the state of the project or program. By definition, the CR is the product of the schedule ratio (SPI) and the cost ratio (CPI). The CR takes into account the impact of schedule and cost. By examining the CR at different times during a project or program, one can determine how the project is developing. There is some qualitative element in interpreting the CR. It is not a cut-and-dry measure that can readily determine a project manager's next move. With experience and a set of guidelines outlined in Exhibit TN2, a project manager may better navigate through a project's natural oscillation around its baseline.

By analyzing Exhibit TN3: EVA Combined Projects, we see why Terry Baker thought the program was on target. The CR for the combined projects shows that the ratio starts off in May at 1.12 but falls to 0.96 in June. A value greater than 1.0 is favorable and indicates that one or both aspects are below cost or ahead of schedule. However, in July the ratio jumps back to 1.02, then to 1.03 in August, and finally to 1.09 in September. The results of combining two projects in an EVA can be deceptive, as is readily depicted here.

By analyzing Exhibit TN4: EVA Software Customization Project, we see the overwhelming effect of having a project way ahead in schedule and way below cost. By examining the CR for the software customization project in detail, we see that the ratio starts off in May at 1.24 but falls to 1.14 in June. Again, a value greater than 1.0 is favorable and indicates that one or both aspects are below cost or ahead of schedule. However, in July the ratio jumps back to 1.28, then to 1.27 in August, and finally to 1.35 in September. The lesson here is that a project that is far ahead of schedule and far below cost can lead to a deceptive interpretation of a combined EVA analysis when the other project is behind schedule and above cost and each project is on the critical path.

By analyzing Exhibit TN5: EVA Technical Infrastructure Project, we see where the problem is. By examining the CR for the technical infrastructure project in detail, we see that the ratio starts off in May at 1.0 but falls to 0.80 in June. A value less than 1.0 is not favorable and indicates that one or both aspects are above cost or behind schedule. However, in July the project recovers and the ratio jumps back to 0.81, then to 0.82 in August, and finally to 0.87 in September. **Exhibit TN9** outlines the first opportunity that the project manager had to begin investigating and potentially make some changes to the project in order to get it back on track for an on-time completion. This point is made even clearer when one compares the software customization project side-by-side with the technical infrastructure project. With excess funds from the under-budget software customization project, the project manager had the option of diverting funds from that project to the technical infrastructure project. By bringing in expert high-performance help where appropriate, he could have helped bring the project back on schedule much sooner.

Case Question Solutions

The assignment can be completed by answering the following questions:

1. Which of the two components are underperforming according to the plan? How do you know this?

The data provided in Exhibits 5, 6, and 7 and the EVA template in Exhibit 8 from the case indicate that the technical infrastructure project is underperforming according to the plan. You know this by looking at the schedule impact for each project. The September SPI for the technical infrastructure project was 94.37%, compared with 113.20% for the software customization project.

2. Are the components of the project within budget? How do you know?

Again using the EVA template and the data from the exhibits, the software customization portion of the project is within budget. You know this by looking at the cost impact for each project. The September CPI for the technical infrastructure project was 91.74%, compared with 118.82% for the software customization project, and is thus over budget.

3. What can you conclude by looking at the combined earned value data for the project?

You can conclude that the overall project was ahead of schedule (SPI = 103.79%) and under budget (CPI = 104.76%). The control ratio, the product of the SPI and the CPI, was 1.09.

4. Why did Terry Baker think that the project was going according to plan the entire time?

Terry Baker most likely believed the program was going according to plan because she was looking at the combined earned value data for both projects. EVA is a powerful tool but can be deceptive when earned value is rolled up for two or more projects. Rolling up separate data from different projects can make projects that are ahead of schedule and under budget compensate for projects that are behind schedule and over budget. The result is an inaccurate picture of the program status when there are two critical paths.

5. How much longer will the project take?

The technical infrastructure project is lagging behind the software customization project and both projects are on the critical path. By reexamining the schedule impact in September for the technical infrastructure project, you can see that the SPI is 94.37%. By subtracting this number from 100%, you determine that the project is 5.63% behind. Thus, based on a six-month or 24-week project, it will take 5.63% × 24 weeks, or approximately 1.35 additional weeks, to complete the project.

6. What should Martin have done earlier in the project timeline to prevent delays?

Martin should have performed an EVA for each portion of the program and independently monitored each project's EVA and CR for possible problems or delays. He could have then made changes as appropriate for each portion of the project to keep it on schedule. If Martin had done this, he could have also reallocated the excess budgeted money from the software project to the infrastructure project to keep the entire program on schedule.

Specifically, Martin could have moved high-performing people who were up to speed on the project and had transferable skill sets from noncritical tasks to critical tasks. He could have also brought in expert high-performance help where appropriate and given them time to get up to speed. Beware of the mythical man-month concept when applying this technique.[2] Finally, Martin could have rebaselined the project with executive buy-in by briefing management on the true project status.

[2] Frederick P. Brooks, *The Mythical Man-Month: Essays on Software Engineering* (Reading, MA: Addison-Wesley, 1975).

7. What should Martin do when managing future projects to prevent similar problems from developing?

To prevent similar problems from occurring when he is managing projects in the future, he should utilize EVA, track the control ratio, and not simply rely on Gantt charts. Martin needs to "get under the hood" and examine each component of the program. In addition, he needs to fully understand EVA so he knows what is wrong with the project, and look at the critical paths so that he understands the implications of fixing the parts most in need of repair.

Teaching the Case

COMMON STUDENT APPROACHES TO THE SOLUTION

All the student groups should be able to fill out the EVA template exhibits and calculate the various ratios quite easily. Based on the results of the EVA spreadsheet template, the students will most likely identify the technical infrastructure project as the source of the delay. Their recommendations will probably advise the project manager to divert funds from the software customization project and hire additional people to put the technical infrastructure project back on track. This is a simplification of the problem and can actually delay the project further. If the task delaying a project involves complex interrelationships, the communication effort can quickly outweigh the decrease in individual task time brought by partitioning. Adding more individuals lengthens, not shortens, the project completion time. This is akin to throwing gasoline on a fire, making the situation worse, not better. More fire requires more gasoline, and thus begins a regenerative cycle that ends in disaster. This concept is better known as the mythical man-month, first identified by Frederick Brooks in the 1960s.[3]

TEACHING APPROACH

Students discuss the case in groups and may be asked to prepare one or two slides to hypothetically present to Christopher Martin. In the MBA class at Kellogg, one team is usually selected to present the case. Students are encouraged to question the presenting team as if they were Christopher Martin. This can make for an interesting dialogue, with students enjoying their roles. Following the group presentation, the instructor facilitates a discussion and presents an optimal solution. In the Kellogg executive program, time is more limited, so the debrief consists of a thirty-minute facilitated discussion.

Next, turn the class discussion over to the case questions and walk through the step-by-step solutions in the EVA spreadsheet template as detailed in the analysis portion of this teaching note. The calculations for the rolling ratios, SPI, CPI, and CR, are pretty straightforward and the results the students get should be more or less consistent with those provided in this note. Verify that the entire class understands the process and solutions before proceeding further.

At this point, you could ask the class to list their recommendations to prevent similar problems from occurring in future projects. In reviewing the recommendations, stress to the students that managing the critical path and fixing the parts most in need of repair is the key to success in this case. Explain that simply working additional overtime to get back on track can

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[3] Brooks, *The Mythical Man-Month.*

exhaust the team. Groups will culturally adapt; for example, in a 12-hour work day, they may spend three hours at lunch and several hours on non-work related tasks. Hence, performance will drop significantly while costs increase.

Knowing when to walk away is another option that each project manager should have in his or her toolkit. Reevaluating the project and thinking through the downside if the project continues to underperform is important. Technology changes quickly and user requirements do as well. If the project drags on indefinitely, the final product may be outdated by the time it is completed. Reducing the scope of the project is yet another option that should be considered.

If the project is deemed critical and must go forward, the project manager must fully understand the EVA so that he or she knows what is wrong. Point out to the students that rebaselining the project with executive buy-in is a good strategy to get back on track. If cost savings are required, staff on tasks that are delayed should be furloughed. This may require renegotiating with a consulting firm that provides the staff, and is more easily accomplished when there is a good, close relationship between the company and the consulting firm. Moving high-performing individuals who have transferable skill sets from noncritical to critical tasks is another way to decrease the delay of a project. Finally, paying bonus incentives to high-performing employees for their best effort, regardless of the project outcome, is another way to trim a project's completion time. All of this, of course, must be done while closely monitoring EVA and the CR to make sure the project stays on track.

Give a final five-minute lecture on what students should do if they find themselves managing a project faced with numerous difficulties. One key point is the variance in skill level of IT staff. When feasible and useful, project managers should consider bringing in experts, because they can be as much as fourteen times more effective than an average person. Be sure to mention that project managers can very rarely simply reassign people from one project to another. The skills and experiences required are often not transferable, and reassignment often causes increased project delay. In the case of Ariba/MED-X, individuals cannot be simply transferred from the software customization project to the infrastructure project. In all likelihood, if consultants are used, the extra software customization personnel will need to be removed and new expert technical infrastructure personnel brought in.

Final debrief slides for use in the debrief lecture are presented in **Exhibit TN10**.

Summary and Key Takeaways

Provided with this teaching note is a PowerPoint presentation that summarizes the case problems that need to be addressed, the main points of EVA, and the key takeaways of the case. These slides may be used to reiterate the advantages of EVA to program management and allow the manager to go under the hood to truly the determine the status of a program.

The key learning of this case is that project analysis tools like Microsoft Project are not project managers. EVA provides earlier detection of project problems and can help the project manager react faster to delays. If a project is delayed, EVA can also answer questions from CFOs about how much the project is going to cost and how late it will be by calculating the estimate at complete.

Another key takeaway is that projects are relatively easy to analyze using EVA. Data can be quickly exported to Microsoft Excel and key parameters, such as the critical ratio, can be quickly scanned to determine if any projects are underperforming. High-level executives can use EVA to track hundreds of projects simultaneously. A key note of caution: although EVA is relatively easy to use, one must be mindful of what one is analyzing. Rolling up several projects into a program and then analyzing them can hide projects in trouble. The resulting rollup is a program roughly on schedule, providing a false sense of security that can come back to haunt managers. Someone should always analyze each project separately to ensure it is doing well on its own. Rollups should be left for higher-level executives who simply want a high-level overview.

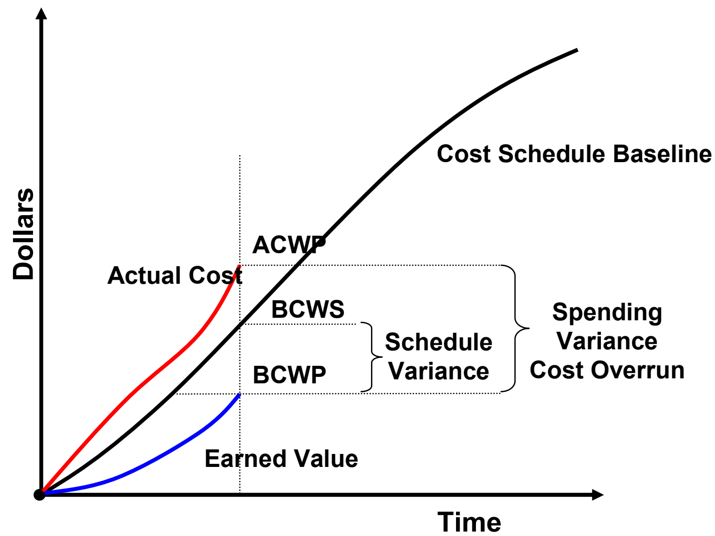
Finally, not all organizations are set up to use EVA. An organization's employees should track their time for each project they work on in order to provide managers the necessary data to perform EVA. A Web-based time reporting system can be implemented to solve this problem, or a simple paper-based system will work as well. An important aspect of EVA is that the analysis is only as good as the data. Employees must understand that accurate and regular time tracking is essential for the successful use of EVA.

Exhibit TN1: Earned Value Analysis

Earned Value

Cost or schedule variances fail to consider the actual progress made on a project or task.

If expenditures are higher or lower than planned it may be bad or good depending on whether progress is in line with that amount of expenditure.



This project is …

Behind Delivery Schedule

Over Budget on Time Schedule
Over Budget on Delivery Schedule

Source: PMI PMBOK

Exhibit TN2: Critical Ratio and Control Limits

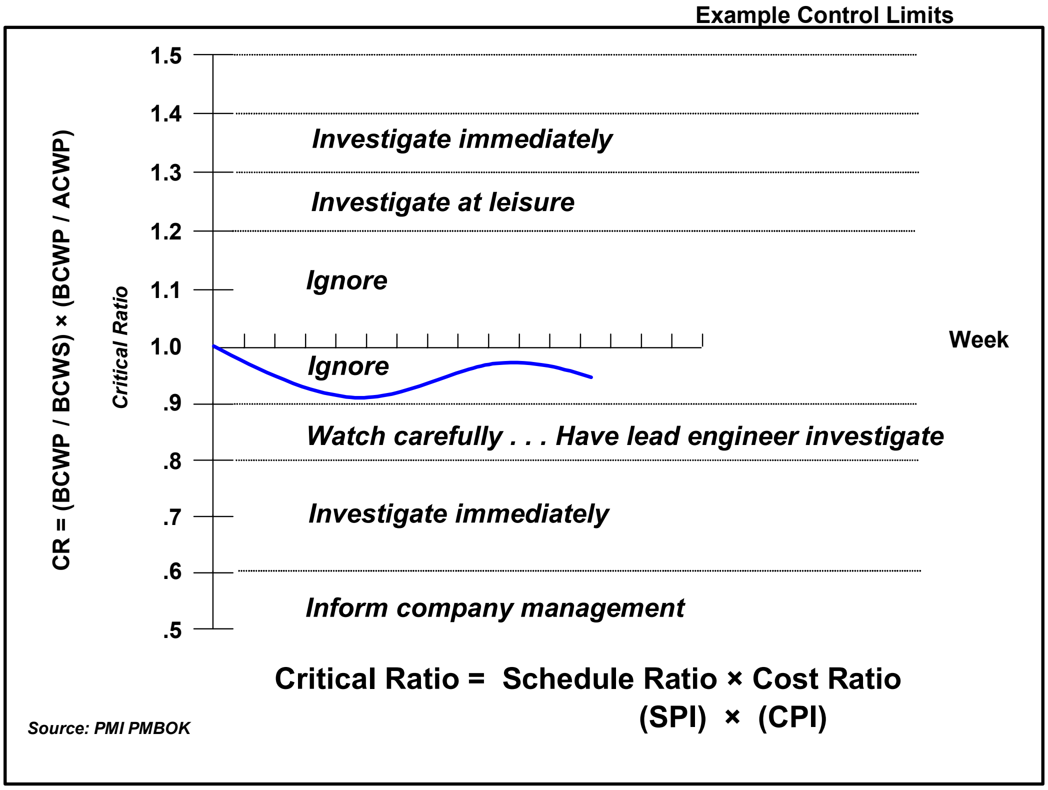


Exhibit TN3: EVA Combined Projects

			May	June	July	Aug	Sep	Oct
Combined Projects								
Monthly Status	**Plan**	**BCWS**	$240,000	$384,000	$384,000	$384,000	$384,000	$120,000
	Actual Burn	**ACWP**	$239,000	$402,000	$357,000	$405,500	$356,000	
	Actual Perform	**BCWP**	$253,250	$367,000	$393,000	$405,000	$425,000	
Rolling Status	**Plan**	**BCWS**	$240,000	$624,000	$1,008,000	$1,392,000	$1,776,000	$1,896,000
	Actual Burn	**ACWP**	$239,000	$641,000	$998,000	$1,403,500	$1,759,500	
	Actual Perform	**BCWP**	$253,250	$620,250	$1,013,250	$1,418,250	$1,843,250	
Rolling Ratios	**Schedule Impact**	**SV = BCWP – BCWS**	$13,250	–$3,750	$5,250	$26,250	$67,250	
		SPI = BCWP / BCWS	105.52%	99.40%	100.52%	101.89%	103.79%	
	Cost Impact	**CV = BCWP – ACWP**	$14,250	–$20,750	$15,250	$14,750	$83,750	
		CPI = BCWP / ACWP	105.96%	96.76%	101.53%	101.05%	104.76%	
	Control Ratio	**CR = SPI × CPI**	1.12	0.96	1.02	1.03	1.09	

Exhibit TN4: EVA Software Customization Project

			May	June	July	Aug	Sep	Oct
Software Customization	**Monthly Plan**							
Monthly Status	**Plan**	**BCWS**	$120,000	$192,000	$192,000	$192,000	$192,000	$60,000
	Actual Burn	**ACWP**	$119,000	$187,000	$165,000	$189,000	$186,000	
	Actual Perform	**BCWP**	$133,250	$197,000	$220,000	$215,000	$240,000	
Rolling Status	**Plan**	**BCWS**	$120,000	$312,000	$504,000	$696,000	$888,000	$948,000
	Actual Burn	**ACWP**	$119,000	$306,000	$471,000	$660,000	$846,000	
	Actual Perform	**BCWP**	$133,250	$330,250	$550,250	$765,250	$1,005,250	
Rolling Ratios	**Schedule Impact**	**SV = BCWP – BCWS**	$13,250	$18,250	$46,250	$69,250	$117,250	
		SPI = BCWP / BCWS	111.04%	105.85%	109.18%	109.95%	113.20%	
	Cost Impact	**CV = BCWP – ACWP**	$14,250	$24,250	$79,250	$105,250	$159,250	
		CPI = BCWP / ACWP	111.97%	107.92%	116.83%	115.95%	118.82%	
	Control Ratio	**CR = SPI × CPI**	1.24	1.14	1.28	1.27	1.35	

Exhibit TN5: EVA Technical Infrastructure Project

			May	June	July	Aug	Sep	Oct
Technical Infrastructure								
Monthly Status	**Plan**	**BCWS**	$120,000	$192,000	$192,000	$192,000	$192,000	$60,000
	Actual Burn	**ACWP**	$120,000	$215,000	$192,000	$216,500	$170,000	
	Actual Perform	**BCWP**	$120,000	$170,000	$173,000	$190,000	$185,000	
Rolling Status	**Plan**	**BCWS**	$120,000	$312,000	$504,000	$696,000	$888,000	$948,000
	Actual Burn	**ACWP**	$120,000	$335,000	$527,000	$743,500	$913,500	
	Actual Perform	**BCWP**	$120,000	$290,000	$463,000	$653,000	$838,000	
Rolling Ratios	**Schedule Impact**	**SV = BCWP – BCWS**	$0	–$22,000	–$41,000	–$43,000	–$50,000	
		SPI = BCWP / BCWS	100.00%	92.95%	91.87%	93.82%	94.37%	
	Cost Impact	**CV = BCWP – ACWP**	$0	–$45,000	–$64,000	–$90,500	–$75,500	
		CPI = BCWP / ACWP	100.00%	86.57%	87.86%	87.83%	91.74%	
	Control Ratio	**CR = SPI × CPI**	1.0	0.80	0.81	0.82	0.87	

Exhibit TN6: Spending on EVA Combined Projects

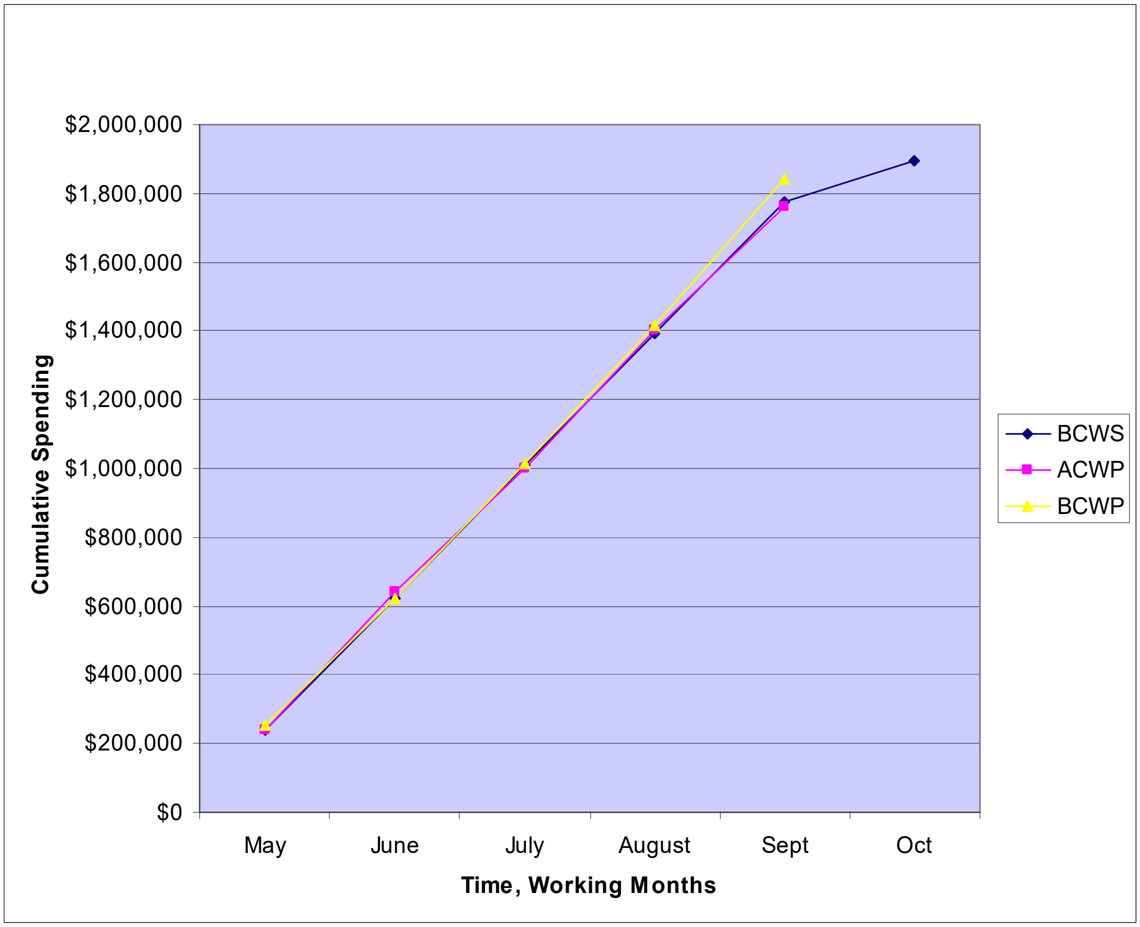


Exhibit TN7: Spending on EVA Software Customization Project

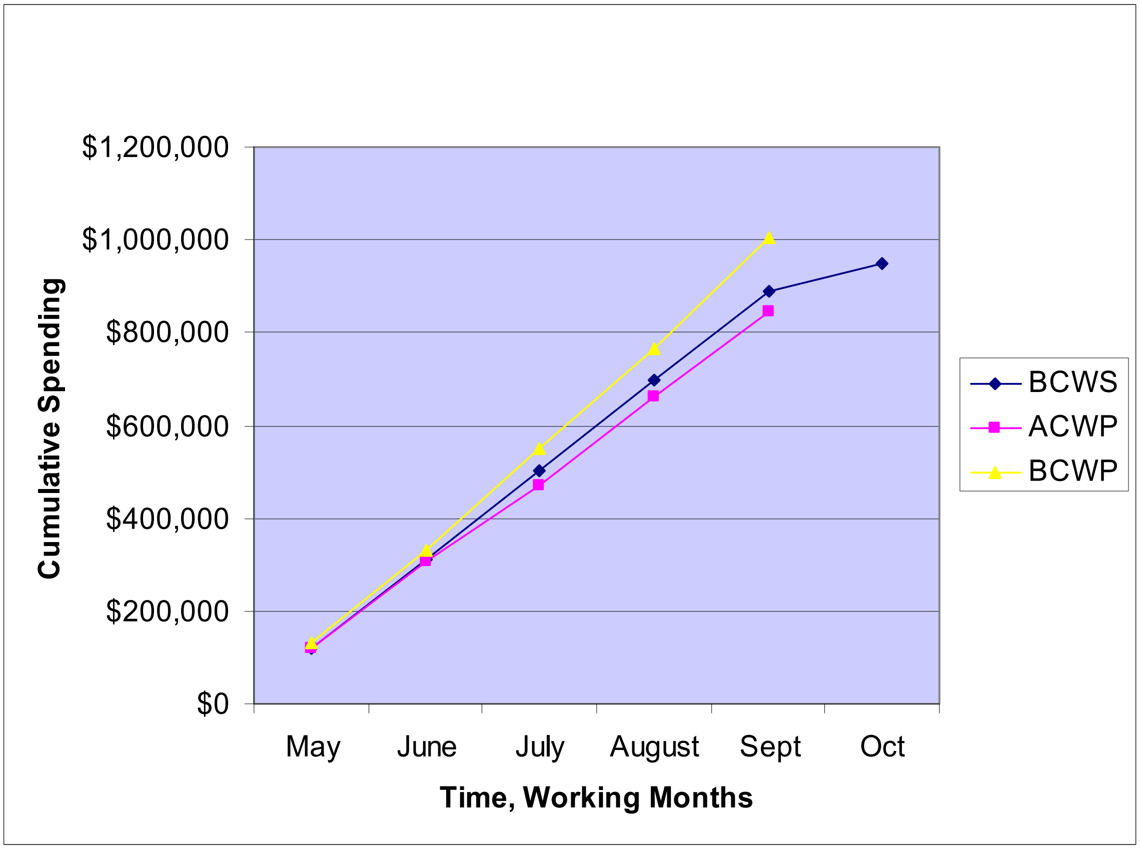


Exhibit TN8: Spending on EVA Technical Infrastructure Project

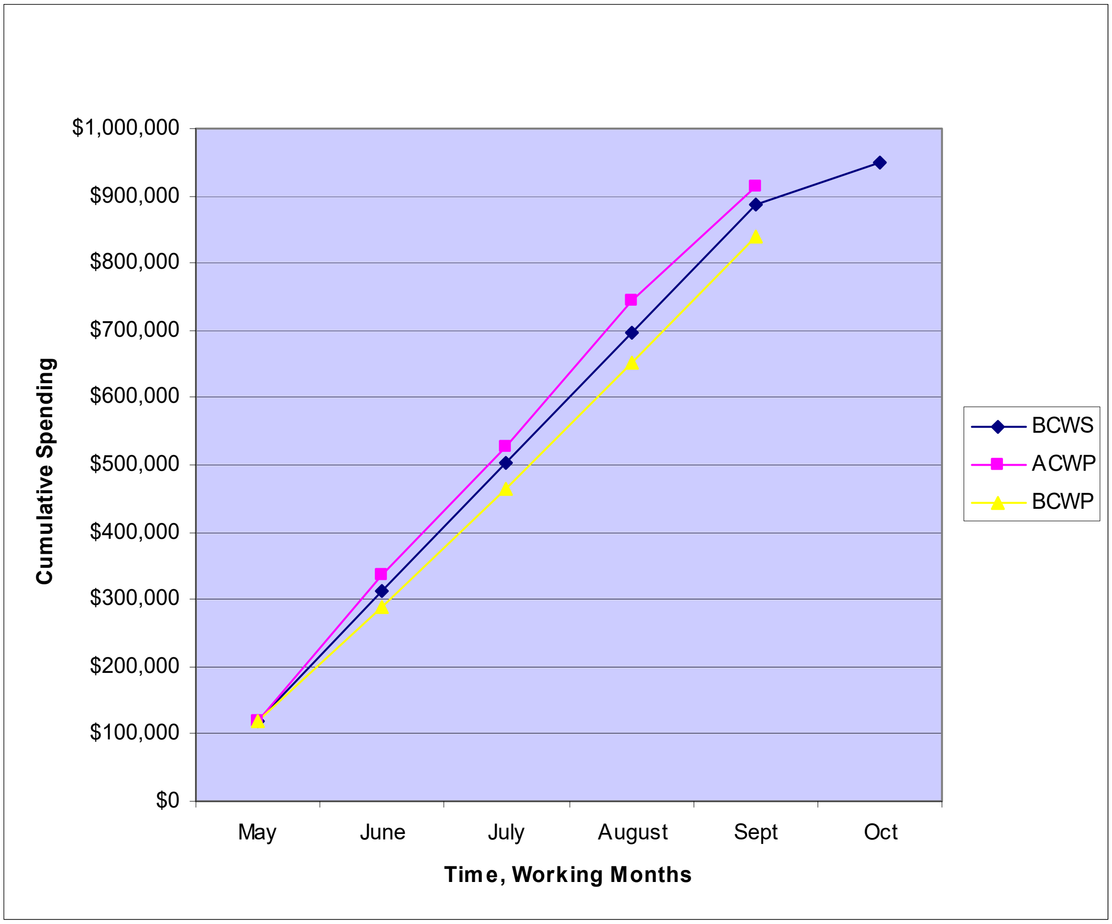


Exhibit TN9: Ariba Med-X Case Earned Value Analysis

			May	June	July	Aug	Sep	Oct
Technical Infrastructure								
Monthly Status	**Plan**	**BCWS**	$120,000	$192,000	$192,000	$192,000	$192,000	$60,000
	Actual Burn	**ACWP**	$120,000	$215,000	$192,000	$216,500	$170,000	
	Actual Perform	**BCWP**	$120,000	$170,000	$173,000	$190,000	$185,000	
Rolling Status	**Plan**	**BCWS**	$120,000	$312,000	$504,000	$696,000	$888,000	$948,000
	Actual Burn	**ACWP**	$120,000	$335,000	$527,000	$743,500	$913,500	
	Actual Perform	**BCWP**	$120,000	$290,000	$463,000	$653,000	$838,000	
Rolling Ratios	**Schedule Impact**	**SV = BCWP – BCWS**	$0	–$22,000	–$41,000	–$43,000	–$50,000	
		SPI = BCWP / BCWS	100.00%	92.95%	91.87%	93.82%	94.37%	
	Cost Impact	**CV = BCWP – ACWP**	$0	–$45,000	–$64,000	–$90,500	–$75,500	
		CPI = BCWP / ACWP	100.00%	86.57%	87.86%	87.83%	91.74%	
	Control Ratio	**CR = SPI × CPI**	1.0	0.80	0.81	0.82	0.87	

First sign the project is off track

Exhibit 10: Case Debrief Slides

Glossary of Terms

- *Budgeted Cost of Work Scheduled (BCWS):* the planned work for the project
- *Actual Cost of Work Performed (ACWP):* the actual cost of the work done on the project
- *Budgeted Cost of Work Performed (BCWP):* the earned value for the project; the actual dollar value of the work done relative to the original plan
- *Cost Variance (CV):* the actual cost variance relative to earned value

  CV = BCWP – ACWP
- *Cost Index (CPI):* ratio of the actual cost of work performed relative to earned value

  CPI = BCWP / ACWP
- *Schedule Variance (SV):* the earned value schedule variance relative to the original plan

  SV = BCWP – BCWS
- *Schedule Index (SPI):* the ratio of planned work to earned value

  SPI = BCWP / BCWS
- *Control Ratio (CR):* the indicator of the combined impact of schedule and cost variances relative to earned value

  CR = SPI × CPI

Ariba MED-X Case Takeaways

- The case is a simplified program with two critical paths using an example Ariba e-procurement implementation
- The EVA for the program roll-up shows everything is OK
- The program manager needs to go under the hood:
  - The software component is ahead of schedule
  - The hardware component is behind schedule
- CR indicator showed a problem with HW in June with CR = 0.80
- As of September, CR is 0.87, so the HW project has made up some ground
- The CFO question: "How much is it going to cost and how late is going to be?"
  - Time is estimated from the SPI for the HW critical path: 7 working days late
  - Cost can be estimated for both HW and SW. For the entire program, SW will cancel the HW loss so the cost is approximately on plan
- These estimates assume the program manager will *not* take action
  - He/she will act to fix the problem, so the estimates are indicators of what will happen if you do nothing
- Best practice: Use earned value and the CR to track and control programs and projects

Exhibit 10 *(continued)*

What should you do when faced with a project in trouble, a hard deadline & additional requirements?

Some suggestions:

- Tell the truth
  - Brief senior management on the true project status with a plan of action
  - Explain the risks and the range of possible outcomes
- Watch out for the trap—when the project is over budget and behind schedule, indiscriminately adding more people will most likely make things worse
- Working 12-hour days burning overtime to get back on track will exhaust your team
  - Three weeks maximum at 12 hr/day pace
  - The group will culturally adapt and performance will drop significantly
- Know when to say no and walk away
  - Can you really deliver the new requirements? If no, then say no
  - Decide whether killing the project at this point is best for the company
  - Think through the downside risk if you continue. Give executives options and possible outcomes to make a decision
- *Reduce scope where possible*

If you have to go forward . . .

- Understand the EVA so you know what is wrong with the project and look at the critical paths so you understand the implications of fixing the most broken pieces
- Re-baseline the project with executive buy-in
  - Ranges of costs and delivery dates for a menu of reduced scope
  - Be realistic in your assumptions
- Where possible, furlough resources on tasks that are delayed
- Move high-performing resources (people) who are up to speed on the project and have transferable skill sets from noncritical tasks to critical tasks
  - Allow time for them to get up to speed on the new problem
  - Watch out for tasks with complex interrelationships
- Bring in expert high-performance help where appropriate—allow time for these new people to get up to speed
- Pay bonus incentives to high-performing employees for their best effort, win or lose
- Closely monitor EVA and the CR to make sure the project stays on track

"When one hears of disastrous schedule slip in project, [one] imagines that a series of major calamities must have befallen it. Usually, however, the disaster is due to termites, not tornadoes . . . the schedule slips one day at a time."

—Frederick P. Brooks, *The Mythical Man-Month*